

Roots (Part IV): Negative Radicands, Imaginary Numbers, and Complex Numbers

Vocabulary

Base Power Index Radicand Radical Symbol

3^5 $\sqrt[n]{y}$

Raising a Negative Number to an Even or Odd Power

Even Powers Produce Positive Numbers

$$\text{I) } (-2)^2 = (-2)(-2) = 4 \quad \text{II) } (-2)^4 = (-2)(-2)(-2)(-2) = 16$$

$$\text{III) } (-2)^6 = (-2)(-2)(-2)(-2)(-2)(-2) = 64$$

Odd Powers Produce Negative Numbers, Assuming that the Base is Negative

$$\text{IV) } (-2)^3 = (-2)(-2)(-2) = -8 \quad \text{V) } (-2)^5 = (-2)(-2)(-2)(-2)(-2) = -32$$

$$\text{VI) } (-2)^7 = (-2)(-2)(-2)(-2)(-2)(-2)(-2) = -128$$

Negative Variables under Radical Symbols

If x is not assumed to be positive, then

$\sqrt{x^2} \neq x$, but $\sqrt{x^2} = |x|$, and if the index (n) of a root ($\sqrt[n]{x}$) is even, then there are other examples where absolute value symbols must be used.

Example: Not $\sqrt{(-3)^2} = -3$, but $\sqrt{(-3)^2} = \sqrt{9} = 3$

Simplify. ★ Do not assume that the unknowns are positive.

$$\text{① } \sqrt{x^{14}} = |x^7|$$

$$\text{② } \sqrt[3]{x^9} = x^3$$

$$\text{③ } \sqrt{x^4} = x^2$$

$$\text{④ } \sqrt[5]{x^5} = x$$

$$\text{⑤ } \sqrt[4]{x^4} = |x|$$

$$\text{⑥ } \sqrt[4]{x^{16}} = x^4$$

$$\text{⑦ } \sqrt{x^{22}} = |x^{11}|$$

$$\text{⑧ } \sqrt[3]{x^{15}} = x^5$$

$$\text{⑨ } \sqrt{x^{16}} = x^8$$

$$\text{⑩ } \sqrt[7]{x^{14}} = x^2$$

$$\text{⑪ } \sqrt[4]{x^{20}} = |x^5|$$

$$\text{⑫ } \sqrt[4]{x^8} = x^2$$

$$\text{⑬ } \sqrt{9x^6} = \sqrt{9} \sqrt{x^6} = 3|x^3|$$

$$\text{⑭ } \sqrt{18x^8} = \sqrt{9 \cdot 2 \cdot x^8} = \sqrt{9} \sqrt{2} \sqrt{x^8} = 3 \cdot \sqrt{2} \cdot x^4 = 3x^4\sqrt{2}$$

$$\text{⑮ } \sqrt[4]{256x^6} = \sqrt[4]{256 \cdot x^4 \cdot x^2} = \sqrt[4]{256} \cdot \sqrt[4]{x^4} \cdot \sqrt[4]{x^2} = 4|x|\sqrt[4]{x^2}$$

$$(16) \sqrt[6]{x^{18}} = \boxed{|x^3|}$$

$$(17) \sqrt{49x^{10}} = \sqrt{49} \cdot \sqrt{x^{10}} = \boxed{7|x^5|}$$

$$(18) \sqrt{20x^4} = \sqrt{4 \cdot 5 \cdot x^4} = \sqrt{4} \sqrt{5} \sqrt{x^4} = 2\sqrt{5}x^2 = \boxed{2x^2\sqrt{5}}$$

$$(19) \sqrt[4]{81x^6} = \sqrt[4]{81x^4 \cdot x^2} = \sqrt[4]{81} \sqrt[4]{x^4} \sqrt[4]{x^2} = \boxed{3|x|\sqrt[4]{x^2}}$$

$$(20) \sqrt{x^8} = \boxed{x^4}$$

Imaginary Numbers

$$i = \sqrt{-1}$$

$$i^2 = -1$$

Examples

$$i, -5i, 7i, i\sqrt{3}, \pi i, \frac{1}{2}i$$

Complex Numbers

Real Imaginary

$$a + bi$$

Examples

$$5 + 3i, -4 + i, \sqrt{7} - 2i, -9 - i\sqrt{8}, \frac{3}{4} + \frac{5}{7}i, \pi + 2i, 10, 6i, i$$

Simplify.

$$(21) (5 + 2i) + (-1 + 3i)$$

$$\boxed{4 + 5i}$$

$$(23) (8 - i) + (-6 - 7i)$$

$$\boxed{2 - 8i}$$

$$(25) (-6 + 3i) - (4 - 9i)$$

$$-6 + 3i + (-4) + 9i$$

$$\boxed{-10 + 12i}$$

$$(27) (11 + 8i) - (-7 + i)$$

$$11 + 8i + 7 - i$$

$$\boxed{18 + 7i}$$

$$(29) 3i(4i) = 12i^2 = 12(-1) = \boxed{-12}$$

$$(30) -8(-6i) = \boxed{48i}$$

$$(31) -7i(4i) = -28i^2 = -28(-1) = \boxed{28}$$

$$(32) 5i(3 + 8i) = 5i(3) + 5i(8i) = 15i + 40i^2 = 15i + 40(-1) = \boxed{15i - 40}$$

$$(33) -6i(-4 + 9i) = -6i(-4) - 6i(9i) = 24i - 54i^2 = 24i - 54(-1) = \boxed{24i + 54}$$

$$(22) (-3 - 5i) + (-2 + 4i)$$

$$\boxed{-5 - i}$$

$$(24) (1 + i) + (3 - 10i)$$

$$\boxed{4 - 9i}$$

$$(26) (-1 - 4i) - (2 + 5i)$$

$$-1 - 4i + (-2) - 5i$$

$$\boxed{-3 - 9i}$$

$$(28) (12 - 6i) - (-7 - 6i)$$

$$12 - 6i + 7 + 6i$$

$$\boxed{19}$$

$$(34) i(-3-10i) = i(-3) + i(-10i) = -3i - 10i^2 = -3i - 10(-1) = \boxed{-3i+10}$$

$$(35) 5i(7) = \boxed{35i}$$

$$(36) 3i(-2i) = -6i^2 = -6(-1) = \boxed{6}$$

$$(37) (-5i)(-7i) = 35i^2 = 35(-1) = \boxed{-35}$$

$$(38) 7i(2+3i) = 7i(2) + 7i(3i) = 14i + 21i^2 = 14i + 21(-1) = \boxed{14i-21}$$

$$(39) -3i(6-5i) = -3i(6) - 3i(-5i) = -18i + 15i^2 = -18i + 15(-1) = \boxed{-18i-15}$$

$$(40) -i(-4-3i) = -i(-4) - i(-3i) = 4i + 3i^2 = 4i + 3(-1) = \boxed{4i-3}$$

(41) $(4-2i)(5+3i)$ FOIL	(42) $(8-5i)(6-7i)$
$4(5) + 4(3i) - 2i(5) - 2i(3i)$	$8(6) + 8(-7i) - 5i(6) - 5i(-7i)$
$20 + 12i - 10i - 6i^2$	$48 - 56i - 30i + 35i^2$
$20 + 2i - 6(-1)$	$48 - 86i + 35(-1)$
$20 + 2i + 6$	$48 - 86i - 35$
$\boxed{26+2i}$	$\boxed{13-86i}$

$$(43) (-1+i)(-3-2i)$$

$$-1(-3) - 1(-2i) + i(-3) + i(-2i)$$

$$3 + 2i - 3i - 2i^2$$

$$3 - i - 2(-1) = 3 - i + 2 = \boxed{5-i}$$

$$(44) (7-6i)(7-8i) = 7(7) + 7(-8i) - 6i(7) - 6i(-8i)$$

$$= 49 - 56i - 42i + 48i^2$$

$$= 49 - 98i + 48(-1)$$

$$= 49 - 98i - 48$$

$$= \boxed{1-98i}$$

(45) $(-2+3i)(-1+10i)$	(46) $(-4-i)(-3-2i)$
$-2(-1) - 2(10i) + 3i(-1) + 3i(10i)$	$-4(-3) - 4(-2i) - i(-3) - i(-2i)$
$2 - 20i - 3i + 30i^2$	$12 + 8i + 3i + 2i^2$
$2 - 23i + 30(-1)$	$12 + 11i + 2(-1)$
$2 - 23i - 30$	$12 + 11i - 2$
$\boxed{-28-23i}$	$\boxed{10+11i}$

$$(47) (6+4i)^2$$

$$(6+4i)(6+4i)$$

$$36 + 24i + 24i + 16i^2$$

$$36 + 48i - 16$$

$$\boxed{20+48i}$$

$$(48) (2-3i)^2$$

$$(2-3i)(2-3i)$$

$$4 - 6i - 6i + 9i^2$$

$$4 - 12i - 9$$

$$\boxed{-5-12i}$$

Write the following expressions in $a+bi$ form.

$$(49) \frac{2}{3+i}$$

$$(50) \frac{-8}{2-5i}$$

$$(51) \frac{3}{1+2i}$$

$$\frac{2(3-i)}{(3+i)(3-i)}$$

$$\frac{-8(2+5i)}{(2-5i)(2+5i)}$$

$$\frac{3(1-2i)}{(1+2i)(1-2i)}$$

$$\frac{6-2i}{9-3i+3i-i^2}$$

$$\frac{-16-40i}{4+10i-10i-25i^2}$$

$$\frac{3-6i}{1-2i+2i-4i^2}$$

$$\frac{6-2i}{9-(-1)}$$

$$\frac{-16-40i}{4-25(-1)}$$

$$\frac{3-6i}{1-4(-1)}$$

$$\frac{6-2i}{9+1}$$

$$\frac{-16-40i}{4+25}$$

$$\frac{3-6i}{1+4}$$

$$\frac{6-2i}{10}$$

$$\frac{-16-40i}{29}$$

$$\frac{3-6i}{5}$$

$$\frac{6}{10} - \frac{2i}{10}$$

$$-\frac{16}{29} - \frac{40i}{29}$$

$$\frac{3}{5} - \frac{6i}{5}$$

$$\frac{3}{5} - \frac{i}{5}$$

$$\frac{3}{5} - \frac{1i}{5}$$

$$a+bi$$

$$(52) \frac{10}{6-3i}$$

$$(53) \frac{4+i}{-2+i}$$

$$(54) \frac{9-2i}{-3-7i}$$

$$\frac{10(6+3i)}{(6-3i)(6+3i)}$$

$$\frac{(4+i)(-2-i)}{(-2+i)(-2-i)}$$

$$\frac{(9-2i)(-3+7i)}{(-3-7i)(-3+7i)}$$

$$\frac{60+30i}{36+18i-18i-9i^2}$$

$$\frac{-8-4i-2i-i^2}{4+2i-2i-i^2}$$

$$\frac{-27+63i+6i-14i^2}{9-21i+21i-49i^2}$$

$$\frac{60+30i}{36-9i^2}$$

$$\frac{-8-6i-(-1)}{4-(-1)}$$

$$\frac{-27+69i-14(-1)}{9-49i^2}$$

$$\frac{60+30i}{36-9(-1)}$$

$$\frac{-8-6i+1}{4+1}$$

$$\frac{-27+69i+14}{9-49(-1)}$$

$$\frac{60+30i}{36+9}$$

$$\frac{-7-6i}{5}$$

$$\frac{-13+69i}{9+49}$$

$$\frac{60+30i}{45}$$

$$-\frac{7}{5} - \frac{6i}{5}$$

$$\frac{-13+69i}{58}$$

$$\frac{60}{45} + \frac{30i}{45}$$

$$\frac{4}{3} + \frac{2i}{3}$$

$$-\frac{13}{58} + \frac{69i}{58}$$

$$\begin{aligned} (55) \frac{-2+i}{-8+i} &= \frac{(-2+i)(-8-i)}{(-8+i)(-8-i)} = \frac{16+2i-8i-i^2}{64+8i-8i-i^2} = \frac{16-6i-(-1)}{64-(-1)} \\ &= \frac{16-6i+1}{64+1} \\ &= \frac{17-6i}{65} \\ &= \boxed{\frac{17}{65} - \frac{6}{65}i} \end{aligned}$$

$$\begin{aligned} (56) \frac{1+2i}{-2-4i} &= \frac{(1+2i)(-2+4i)}{(-2-4i)(-2+4i)} = \frac{-2+4i-4i+8i^2}{4-8i+8i-16i^2} = \frac{-2+8(-1)}{4-16(-1)} \\ &= \frac{-2-8}{4+16} \\ &= \frac{-10}{20} \\ &= \boxed{-\frac{1}{2}} \end{aligned}$$

$$\begin{aligned} (57) \frac{2-8i}{3i} &= \frac{(2-8i)(i)}{3i(i)} = \frac{2i-8i^2}{3i^2} = \frac{2i-8(-1)}{3(-1)} \\ &= \frac{2i+8}{-3} \end{aligned}$$

$$\begin{aligned} (58) \frac{-7+3i}{i} &= \frac{(-7+3i)(i)}{i(i)} = \boxed{-\frac{8}{3} - \frac{2}{3}i} \\ &= \frac{-7i+3i^2}{i^2} = \frac{-7i+3(-1)}{-1} = \frac{-7i-3}{-1} = \boxed{3+7i} \end{aligned}$$

Powers of the Imaginary Unit

$$i^0 = 1 \quad i^2 = -1 \quad i^3 = i^2 \cdot i = -1 \cdot i = -i$$

$$i^4 = i^3 \cdot i = -i \cdot i = -i^2 = -(-1) = 1$$

$$i^5 = i^4 \cdot i = 1 \cdot i = i$$

For the problems below, use the pattern.

$$i^0, i^2, i^3, i^4, i^5, i^6, i^7, i^8, i^9, i^{10}, i^{11}, i^{12}, i^{13}, i^{14}, i^{15}, i^{16}$$

$$1, -1, -i, 1, i, -1, -i, 1, i, -1, -i, 1, i, -1, -i, 1$$

*Hint: Divide the power by four. If the remainder is zero, then the answer is 1. If the remainder is one, then the answer is i . If the remainder is two, then the answer is -1 . If the remainder is three, then the answer is $-i$.

$$(59) i^{26} = \boxed{-1} \quad (60) i^{35} = \boxed{-i} \quad (61) i^{80} = \boxed{1} \quad (62) i^{20} = \boxed{1}$$

$$\begin{array}{r} 6 \\ 4 \overline{)26} \\ \underline{24} \\ 2 \end{array}$$

$$\begin{array}{r} 8 \\ 4 \overline{)35} \\ \underline{32} \\ 3 \end{array}$$

$$\begin{array}{r} 20 \\ 4 \overline{)80} \\ \underline{80} \\ 0 \end{array}$$

$$\begin{array}{r} 5 \\ 4 \overline{)20} \\ \underline{20} \\ 0 \end{array}$$

$$(63) i^{25} = \boxed{i}$$

$$(64) i^{75} = \boxed{-i}$$

$$\begin{array}{r} 6 \\ 4 \overline{)25} \\ \underline{24} \\ 1 \end{array}$$

$$\begin{array}{r} 18 \\ 4 \overline{)75} \\ \underline{68} \\ 7 \\ \underline{68} \\ 3 \\ \underline{32} \\ 3 \end{array}$$

Write each expression in terms of i .

$$(65) \sqrt{-3} = \sqrt{-1(3)} = \sqrt{-1} \sqrt{3} = \boxed{i\sqrt{3}}$$

$$(66) \sqrt{-81} = \sqrt{-1(81)} = \sqrt{-1} \sqrt{81} = i(9) = \boxed{9i}$$

$$(67) \sqrt{-2} = \boxed{i\sqrt{2}}$$

$$(68) \sqrt{-4} = \boxed{2i}$$

$$(69) \sqrt{-5} = \sqrt{-1(5)} = \sqrt{-1} \sqrt{5} = \boxed{i\sqrt{5}}$$

$$(70) \sqrt{-49} = \sqrt{-1(49)} = \sqrt{-1} \sqrt{49} = i(7) = \boxed{7i}$$

$$(71) \sqrt{-7} = \boxed{i\sqrt{7}}$$

$$(72) \sqrt{-100} = i(10) = \boxed{10i}$$

$$(73) 5\sqrt{-20} + 6\sqrt{-45}$$

$$5\sqrt{-1(4)(5)} + 6\sqrt{-1(9)(5)}$$

$$5\sqrt{-1} \sqrt{4} \sqrt{5} + 6\sqrt{-1} \sqrt{9} \sqrt{5}$$

$$5i \cdot 2 \cdot \sqrt{5} + 6i \cdot 3 \cdot \sqrt{5}$$

$$10i\sqrt{5} + 18i\sqrt{5}$$

$$\boxed{28i\sqrt{5}}$$

$$(74) -3\sqrt{-8} + 2\sqrt{-32}$$

$$-3\sqrt{-1(4)(2)} + 2\sqrt{-1(16)(2)}$$

$$-3\sqrt{-1} \sqrt{4} \sqrt{2} + 2\sqrt{-1} \sqrt{16} \sqrt{2}$$

$$-3i \cdot 2 \cdot \sqrt{2} + 2i \cdot 4 \cdot \sqrt{2}$$

$$-6i\sqrt{2} + 8i\sqrt{2}$$

$$\boxed{2i\sqrt{2}}$$

$$(75) 8\sqrt{-27} - 2\sqrt{-12}$$

$$8\sqrt{-1(9)(3)} - 2\sqrt{-1(4)(3)}$$

$$8\sqrt{-1} \sqrt{9} \sqrt{3} - 2\sqrt{-1} \sqrt{4} \sqrt{3}$$

$$8i \cdot 3 \cdot \sqrt{3} - 2i \cdot 2 \cdot \sqrt{3}$$

$$24i\sqrt{3} - 4i\sqrt{3}$$

$$\boxed{20i\sqrt{3}}$$

$$(76) -\sqrt{-250} + \sqrt{-40}$$

$$-\sqrt{-1(25)(10)} + \sqrt{-1(4)(10)}$$

$$-\sqrt{-1} \sqrt{25} \sqrt{10} + \sqrt{-1} \sqrt{4} \sqrt{10}$$

$$-i \cdot 5 \cdot \sqrt{10} + i \cdot 2 \cdot \sqrt{10}$$

$$-5i\sqrt{10} + 2i\sqrt{10}$$

$$\boxed{-3i\sqrt{10}}$$